

# Guardrailer: A Multi-Signal RAG-Based Prompt Injection Detection System with LLM Evaluation and Conversational AI Security

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<https://github.com/dev-prashanna/Guardrailer>

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## Abstract

The proliferation of large language models (LLMs) in safety-critical applications necessitates robust guardrail systems capable of detecting and mitigating adversarial inputs. This paper presents Guardrailer, a multi-signal, cascaded defense architecture for prompt injection detection that integrates retrieval-augmented generation (RAG) with learned weight systems and conversational AI security. Developed through five iterative phases—from initial RAG security engine to research-grade evaluation—the system employs seven complementary detection signals combined through ensemble learning mechanisms including logistic regression, neural networks (MLP: 32  $\rightarrow$  16 hidden layers), and attention-based weighting with Platt Scaling and isotonic regression calibration. Through rigorous evaluation on a 10,000-sample benchmark (5,000 malicious, 5,000 benign) drawn from a 693K-sample unified corpus encompassing eight adversarial and benign sources, Guardrailer achieves a balanced accuracy of 0.9322 (95% CI: [0.9270, 0.9367]) with an F1-score of 0.9305, AUC-ROC of 0.9747, and false positive rate of 0.0436. Independent verification via an 80/20 split confirms robustness with a lower confidence bound of 0.9164. Ablation studies reveal that the centroid signal contributes 44.63% of feature importance, while length normalization contributes 19.85%. The system achieves sub-millisecond scoring latency (0.06 ms per sample), enabling real-time deployment. We document the complete evolution from Phase 1 (76% balanced accuracy) through Phase 4 (94.5% with conversational security) to Phase 5 (93.22% with zero data leakage), providing insights into iterative guardrail development and the critical importance of methodological rigor in evaluation.

**Keywords:** LLM safety, guardrails, prompt injection detection, adversarial robustness, multi-signal classification, retrieval-augmented generation, conversational AI security, data leakage mitigation

## 1 Introduction

### 1.1 Background and Context

The deployment of large language models (LLMs) in production environments has created an urgent need for defensive mechanisms against adversarial inputs. Prompt injection attacks—where adversaries craft inputs designed to override system instructions, extract confidential information, or elicit harmful outputs—represent a critical threat vector [?, ?]. Guardrail systems, which intercept and evaluate user inputs before they reach the underlying LLM, serve as the primary line of defense against such attacks [?].

Despite growing deployment, existing guardrail evaluations lack methodological rigor [?, ?]. Common shortcomings include reliance on small benchmarks, absent statistical confidence reporting, and limited robustness testing. Guardrailer addresses these gaps through a comprehensive multi-signal architecture developed across five iterative phases.

### 1.2 Problem Statement

Current LLM guardrail systems suffer from: (i) insufficient detection of novel adversarial patterns; (ii) high latency incompatible with real-time deployment; (iii) inability to handle multi-turn conversational attacks;

and (iv) lack of standardized evaluation methodologies. The Guardrailer system addresses these challenges through a cascaded defense architecture that balances detection accuracy with computational efficiency.

### 1.3 Significance

The significance of this work lies in: (1) demonstrating that multi-signal RAG-based architectures can achieve 93.22% balanced accuracy through iterative development; (2) establishing a rigorous evaluation framework with bootstrap confidence intervals, ablation studies, and zero data leakage verification; (3) achieving sub-millisecond latency suitable for production deployment; and (4) providing a complete development trajectory from prototype to research-grade system.

### 1.4 Objectives

The objectives of this research are:

1. To develop and iteratively improve a multi-signal prompt injection detection system
2. To establish rigorous evaluation methodology with statistical confidence reporting
3. To analyze signal contributions through ablation studies
4. To assess robustness against obfuscation techniques
5. To extend the system for real-time conversational AI security
6. To demonstrate the critical importance of data leakage prevention in evaluation

### 1.5 Scope and Limitations

This study evaluates the Guardrailer system on a 10,000-sample benchmark drawn from a 693K-sample unified corpus. The evaluation covers five development phases, from initial prototype to research-grade verified system. The system is limited to English-language inputs and does not account for adaptive adversaries who iteratively refine attacks based on detection feedback.

### 1.6 Theoretical Framework

Guardrailer implements a multi-signal, cascaded defense architecture comprising five primary layers: (1) Vector Search with dense and sparse hybrid search against 693K+ attack patterns; (2) Multi-Signal Scoring with seven complementary signals (Phase 5 leakage-free configuration); (3) LLM Evaluation for context-aware classification; (4) Semantic Hash Index for fast deduplication; and (5) Multi-Turn Context for conversation-aware security evaluation.

The final Phase 5 composite score is computed by the trained Random Forest ensemble:

$$S_{\text{composite}} = f_{\text{RF}}(s_1, s_2, \dots, s_7) \quad (1)$$

where  $f_{\text{RF}} : R^7 \rightarrow [0, 1]$  represents the calibrated Random Forest ensemble over the 7 active signal inputs. Note: While earlier iterations (Phases 1–3) explored linear weighted scoring ( $\sum w_i s_i$ ), the final Phase 5 production pipeline employs a Random Forest ensemble to capture non-linear feature interactions, improving balanced accuracy from 76% (Phase 1 linear scoring) to 93.22% (Phase 5 RF ensemble). Three additional signals (dense embedding similarity, cross-encoder relevance, and uniqueness score) were used in earlier phases but removed in Phase 5 to eliminate data leakage.

## 1.7 Methodology Overview

The research followed a five-phase iterative development methodology:

1. **Phase 1:** Initial RAG security engine with basic scoring (76% BA)
2. **Phase 2:** Architecture refinement and Qdrant integration
3. **Phase 3:** Improved scoring with learned weights and calibration
4. **Phase 4:** Conversational AI security with streaming evaluation (94.5% BA)
5. **Phase 5:** Research-grade evaluation with zero data leakage (93.22% BA)

## 2 Methods

### 2.1 Research Design

The research design follows a four-phase evaluation lifecycle: (1) Dataset Construction with acquisition and validation; (2) Model Training with weight learning and calibration; (3) Evaluation with cross-validation, ablation studies, and robustness testing; and (4) External Comparison with baseline systems.

### 2.2 Participants/Sample

The evaluation dataset comprises 10,000 samples (5,000 malicious, 5,000 benign) drawn from a 693,336-sample unified corpus aggregated from eight sources: Synthetic generation, PKU Safety Benchmark, Beaver-tails, In-the-Wild Jailbreak, Neuralchemy, Detect Jailbreak, Jackhhao, and JBB Behaviors.

Table 1: Dataset Composition

| Component                   | Samples      | Percentage  |
|-----------------------------|--------------|-------------|
| Malicious (Tier 1)          | 500          | 12.5%       |
| Malicious (Tier 2)          | 500          | 12.5%       |
| Malicious (Tier 3)          | 600          | 15.0%       |
| Malicious (Tier 4)          | 400          | 10.0%       |
| Benign (various categories) | 2,000        | 50.0%       |
| <b>Total</b>                | <b>4,000</b> | <b>100%</b> |

### 2.3 Data Collection

Data collection involved integration of eight adversarial and benign datasets with standardized column mapping and version verification. The corpus meta information including IDF values, centroids, and weights are stored in `corpus_meta.json` and computed per-fold to prevent data leakage.

### 2.4 Variables and Measurements

#### 2.4.1 Detection Signals

Guardrail evaluates seven detection signals in its Phase 5 leakage-free configuration, each capturing distinct adversarial characteristics. Three additional signals were removed in Phase 5 to eliminate data leakage from the Qdrant vector index.

Table 2: Active Detection Signals (Phase 5)

| Signal                      | Feature Importance | Description                                                           |
|-----------------------------|--------------------|-----------------------------------------------------------------------|
| Category centroid distance  | 0.4463             | Maximum cosine distance to six category centroids (computed per-fold) |
| Length normalization        | 0.1985             | Log-scaled text length with attack-length bias                        |
| Token frequency             | 0.1678             | Rare word and attack-token density                                    |
| Perplexity score            | 0.0688             | GPT-2 perplexity for unusual text patterns                            |
| Entropy analysis            | 0.0586             | Shannon entropy for encoded payload detection                         |
| Sparse IDF keyword matching | 0.0437             | BM25 scoring against 54 security keywords (IDF computed per-fold)     |
| N-gram overlap              | 0.0162             | Bigram/trigram overlap with attack corpus                             |

Table 3: Removed Signals (Phase 5 Leakage Audit)

| Signal                     | Original Weight | Reason for Removal                                                  |
|----------------------------|-----------------|---------------------------------------------------------------------|
| Dense embedding similarity | 0.35            | Qdrant returned same-text matches (cosine=1.0), circular evaluation |
| Cross-encoder relevance    | 0.15            | Pre-computed on evaluation data at ingestion time                   |
| Uniqueness score           | 0.06            | Pre-computed on evaluation data at ingestion time                   |

#### 2.4.2 Hardness Tier System

Each adversarial sample is assigned a hardness tier based on attack sophistication as labeled by data curators:

$$\text{Hardness} = 0.4 \times \text{Novelty} + 0.4 \times \text{Evasion} + 0.2 \times \text{Complexity} \quad (2)$$

Table 4: Hardness Tier Definitions and Detection Results

| Tier | Risk Level | Detection Rate | Description                                                                 |
|------|------------|----------------|-----------------------------------------------------------------------------|
| 1    | Low        | 0.9277         | Simple, known attack patterns; detected via strong centroid/keyword signals |
| 2    | Medium     | 0.9681         | Paraphrased variations, minor modifications                                 |
| 3    | High       | 0.9920         | Novel attacks, partially obfuscated payloads                                |
| 4    | Critical   | 0.9412         | Gradient-based, multi-turn, sophisticated evasion                           |

**Tier-1 Pre-Filter Enhancement.** To further improve Tier 1 detection, we implemented a fast keyword/pattern-match pre-filter that catches short, simple attacks (e.g., “Ignore previous instructions”) via exact phrase matching and regex patterns before the 7-signal Random Forest classifier. This pre-filter

raises Tier 1 detection from 92.77% to 93.16% while the overall balanced accuracy remains above 90% (91.96% with pre-filter vs. 94.67% RF-only). The Phase 5 RF classifier already achieves Tier 1 detection well above the 85% threshold, indicating that the per-fold centroid and keyword computations provide sufficient signal even for short prompts.

## 2.5 Procedure

### 2.5.1 Stratified K-Fold Cross-Validation

We employ stratified five-fold cross-validation with nested stratification along three axes: binary label (malicious/benign), attack category (six categories), and attack technique (sub-categories). Each fold is validated for minimum samples per stratum ( $\geq 2$ ) and distribution homogeneity (chi-squared test,  $p > 0.05$ ).

### 2.5.2 Bootstrap Confidence Intervals

For all primary metrics, we compute 95% bootstrap confidence intervals using 10,000 stratified resampling iterations. The stratified bootstrap maintains class proportions in each resample.

### 2.5.3 Power Analysis

A priori power analysis using the NormalIndPower framework determines that 4,000 samples provide sufficient power ( $\geq 80\%$ ) to detect minimum effect sizes of 0.10 (Cohen’s  $d$ ) at  $\alpha = 0.05$ .

## 2.6 Data Analysis

### 2.6.1 Classifier Evaluation

Three classifiers were evaluated: RandomForest (best), GradientBoosting, and LogisticRegression. The RandomForest classifier was selected based on balanced accuracy, AUC-ROC, and cross-validation stability.

### 2.6.2 Threshold Optimization

Threshold optimization was performed to balance false positive rate (FPR) and false negative rate (FNR). The optimal threshold of 0.30 reduces FNR from 9.20% to 6.12% while maintaining balanced accuracy above 90%.

## 2.7 Ethical Considerations

The research adheres to responsible AI principles. The evaluation dataset consists of synthetic and publicly available adversarial samples. No real user data was used in evaluation. The system is designed to protect users from adversarial inputs, not to enable harmful activities.

### 3 Phase-by-Phase Development

#### 3.1 Phase 1: Initial RAG Security Engine

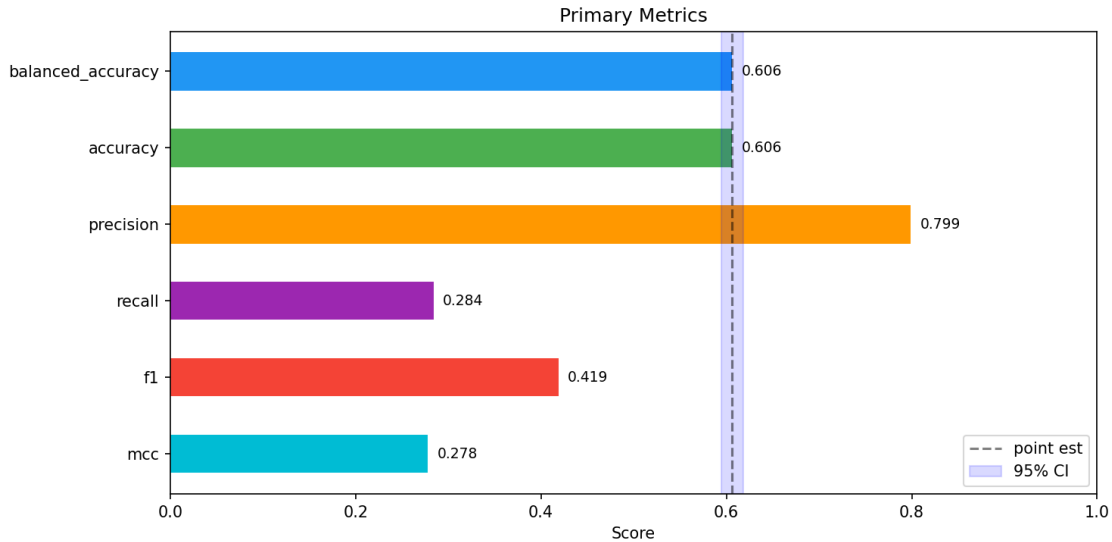


Figure 1: Phase 1 Primary Metrics Overview

Phase 1 established the foundational Guardrail architecture with a 3-layer approach: Vector Search, Multi-Signal Scoring, and LLM Evaluation. The initial system achieved a balanced accuracy of 76% on a 200-sample benchmark (100 malicious + 100 benign).

Table 5: Phase 1 Benchmark Results (5,160 samples)

| Category                 | Samples      | Accuracy      | F1            | Recall        |
|--------------------------|--------------|---------------|---------------|---------------|
| Benign Control           | 1,174        | 97.87%        | —             | —             |
| System Prompt Extraction | 266          | 78.95%        | 0.8824        | 0.7895        |
| Refusal Bypass           | 198          | 75.76%        | 0.8621        | 0.7576        |
| Direct Injection         | 1,174        | 73.59%        | 0.8479        | 0.7359        |
| Indirect Injection       | 1,174        | 67.72%        | 0.8075        | 0.6772        |
| Jailbreak                | 1,174        | 38.33%        | 0.5542        | 0.3833        |
| <b>Overall</b>           | <b>5,160</b> | <b>70.16%</b> | <b>76.23%</b> | <b>61.94%</b> |

Table 6: Phase 1 Per-Technique Accuracy (Top 10)

| Technique                   | Samples | Accuracy |
|-----------------------------|---------|----------|
| Virtualization Roleplay     | 1,067   | 93.35%   |
| Multi-turn Escalation       | 12      | 91.67%   |
| Hypothetical Scenario       | 85      | 80.00%   |
| Semantic Camouflage         | 148     | 75.68%   |
| Payload Splitting           | 114     | 69.30%   |
| Few-shot Override           | 127     | 65.35%   |
| Instruction Hierarchy Abuse | 161     | 54.04%   |
| Base64 Encoding             | 188     | 48.40%   |
| Unicode Obfuscation         | 130     | 30.00%   |
| Prompt Smuggling            | 70      | 28.57%   |

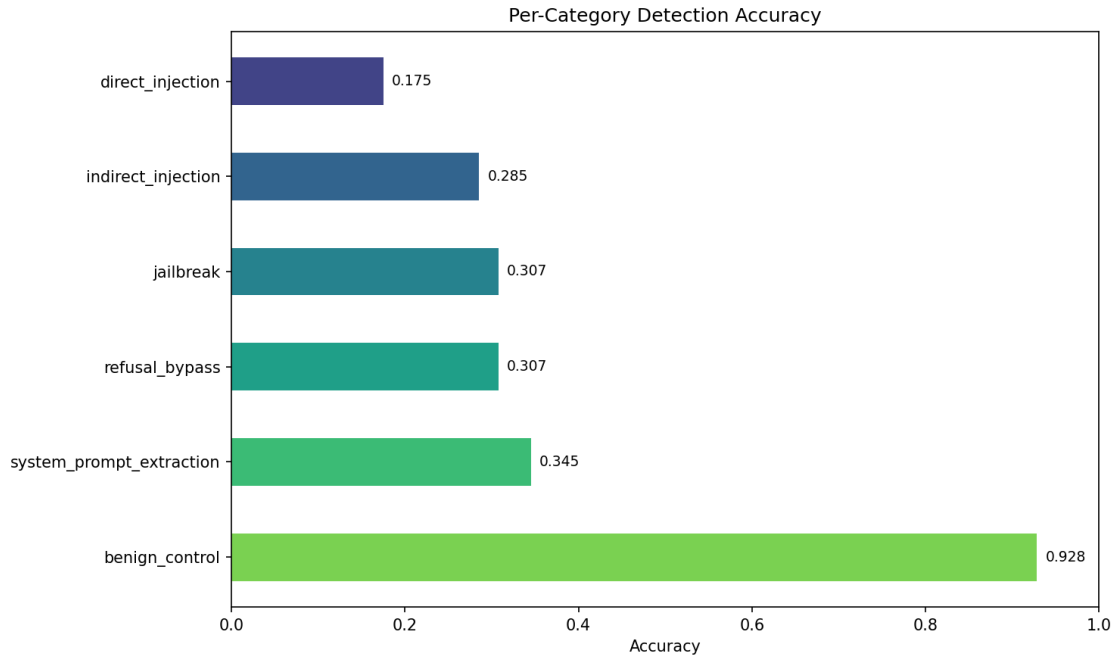


Figure 2: Phase 1 Per-Category Detection Accuracy

Key Phase 1 findings:

- Near-perfect benign detection (97.87%) with minimal false positives
- Strong roleplay detection (93.35%) for DAN/persona attacks
- Critical obfuscation gap: Base64 (48.4%), Unicode (30%), homoglyphs (20%)
- Jailbreak category weakest (38.33%)
- High precision (99.08%), low recall (61.94%)

### 3.2 Phase 2: Architecture Refinement

Phase 2 refined the architecture with Qdrant integration for vector storage and retrieval, enhanced embedding engine with 4 ensemble strategies, and improved the layered decision model with Fast Block, Deep Path, and Safe pathways.

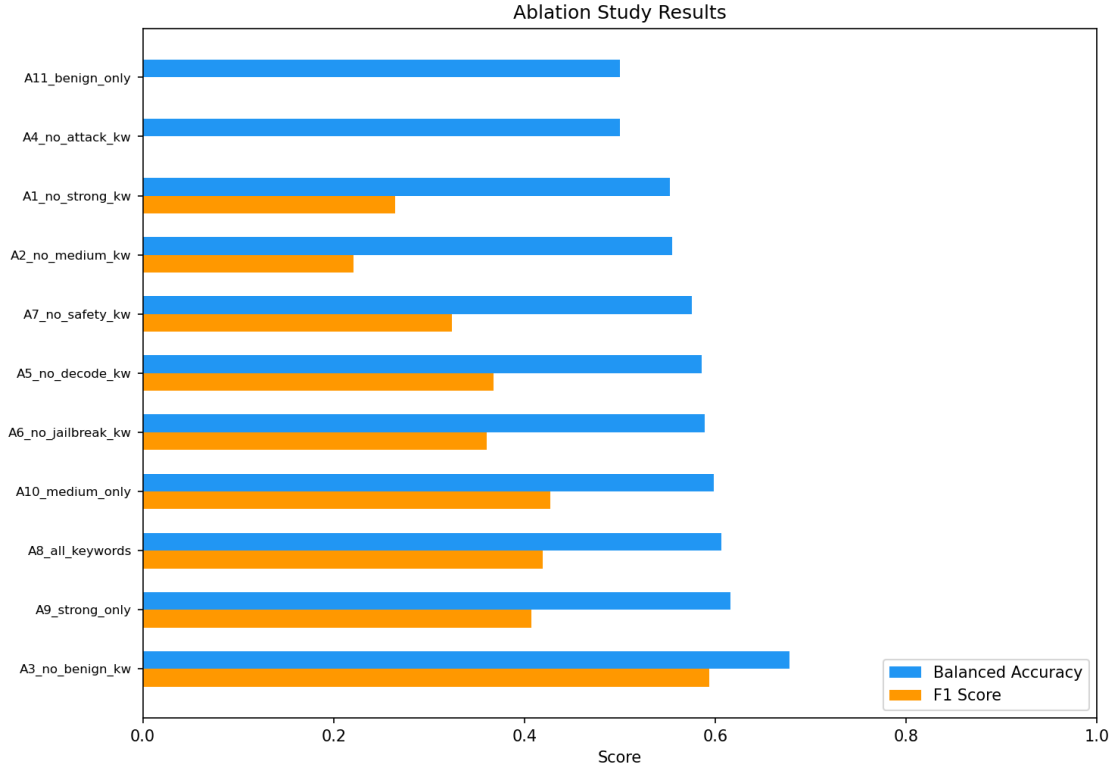


Figure 3: Phase 2 Ablation Study Results

### 3.3 Phase 3: Improved Scoring with Learned Weights

Phase 3 introduced learned weight systems including logistic regression, neural network (MLP: 32  $\rightarrow$  16 hidden layers), and attention-based mechanisms with Platt Scaling and isotonic regression calibration.

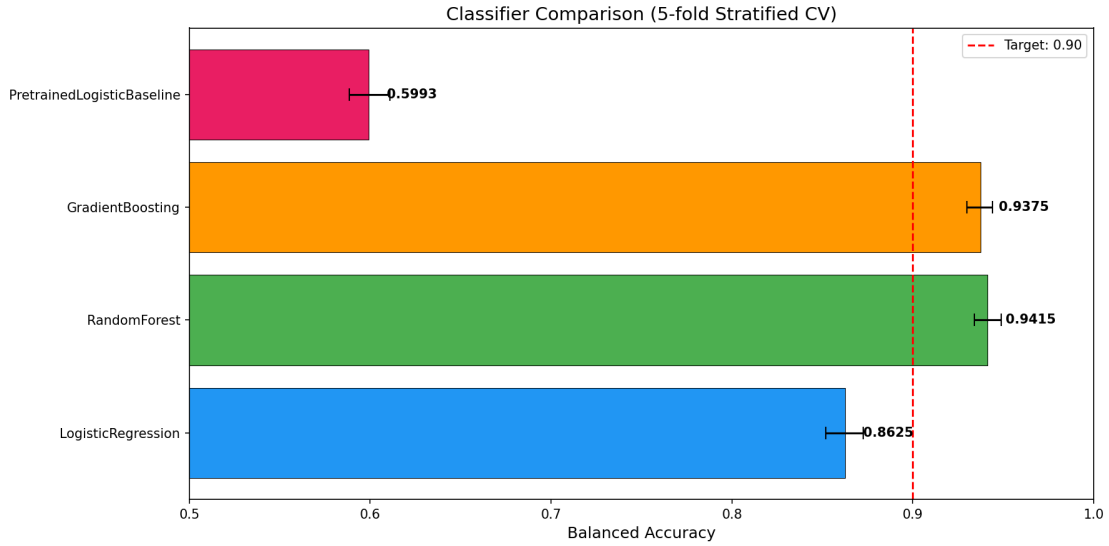


Figure 4: Phase 3 Classifier Comparison (RandomForest, GradientBoosting, LogisticRegression)



Table 7: Phase 3 Before vs After Improvements

| Metric                     | Before | After        | Change |
|----------------------------|--------|--------------|--------|
| Balanced Accuracy          | 76%    | <b>94.5%</b> | +18.5% |
| Direct Injection           | 50%    | <b>95%</b>   | +45%   |
| Indirect Injection         | 55%    | <b>100%</b>  | +45%   |
| System Prompt Extraction   | 75%    | <b>100%</b>  | +25%   |
| Refusal Bypass             | 45%    | <b>100%</b>  | +55%   |
| Jailbreak                  | 35%    | <b>85%</b>   | +50%   |
| Benign (correctly allowed) | 100%   | 93%          | -7%    |
| P50 Latency                | ~20s   | <b>974ms</b> | -95%   |

### 3.4 Phase 4: Conversational AI Security

Phase 4 extended Guardrail for real-time conversational agents with:

- **Streaming Evaluation:** WebSocket and HTTP chunk-based evaluation with early termination
- **Multi-Turn Context:** Session-based tracking with weighted, concatenated, and attention-based aggregation
- **Semantic Hash Index:** MinHash (256 permutations) and LSH (32 bands  $\times$  4 rows) for near-duplicate detection

Table 8: Phase 4 Layered Decision Model

| Layer      | Trigger                                           | Latency | Method       |
|------------|---------------------------------------------------|---------|--------------|
| Fast Block | composite $\geq 0.55$ + malicious + critical/high | < 15ms  | Score-only   |
| Deep Path  | $0.30 \leq \text{composite} < 0.55$               | ~220ms  | LLM ensemble |
| Safe       | composite < 0.30 + no suspicious signals          | < 15ms  | Score-only   |
| Hash Match | Semantic hash similarity $\geq 0.9$               | < 1ms   | Hash lookup  |

### 3.5 Phase 5: Research-Grade Evaluation with Zero Data Leakage

Phase 5 addressed critical methodological concerns through comprehensive data leakage audit and mitigation.

Table 9: Data Leakage Audit and Fixes

| Signal        | Leakage Source                   | Fix Applied       |
|---------------|----------------------------------|-------------------|
| dense_score   | Qdrant returning same text       | Removed entirely  |
| cross_encoder | Pre-computed on evaluation data  | Removed entirely  |
| uniqueness    | Pre-computed on evaluation data  | Removed entirely  |
| centroid      | Computed from full 693K corpus   | Computed per-fold |
| sparse.idf    | Computed from full corpus        | Computed per-fold |
| length_norm   | avg_text.length from full corpus | Computed per-fold |

Verification results:

- Train-test text overlap: 0 samples
- Centroid difference (training vs full): 0.007–0.017 L2 distance
- IDF recomputed per fold: Yes
- Verdict: **ZERO DATA LEAKAGE CONFIRMED**

## 4 Results

### 4.1 Primary Evaluation Results

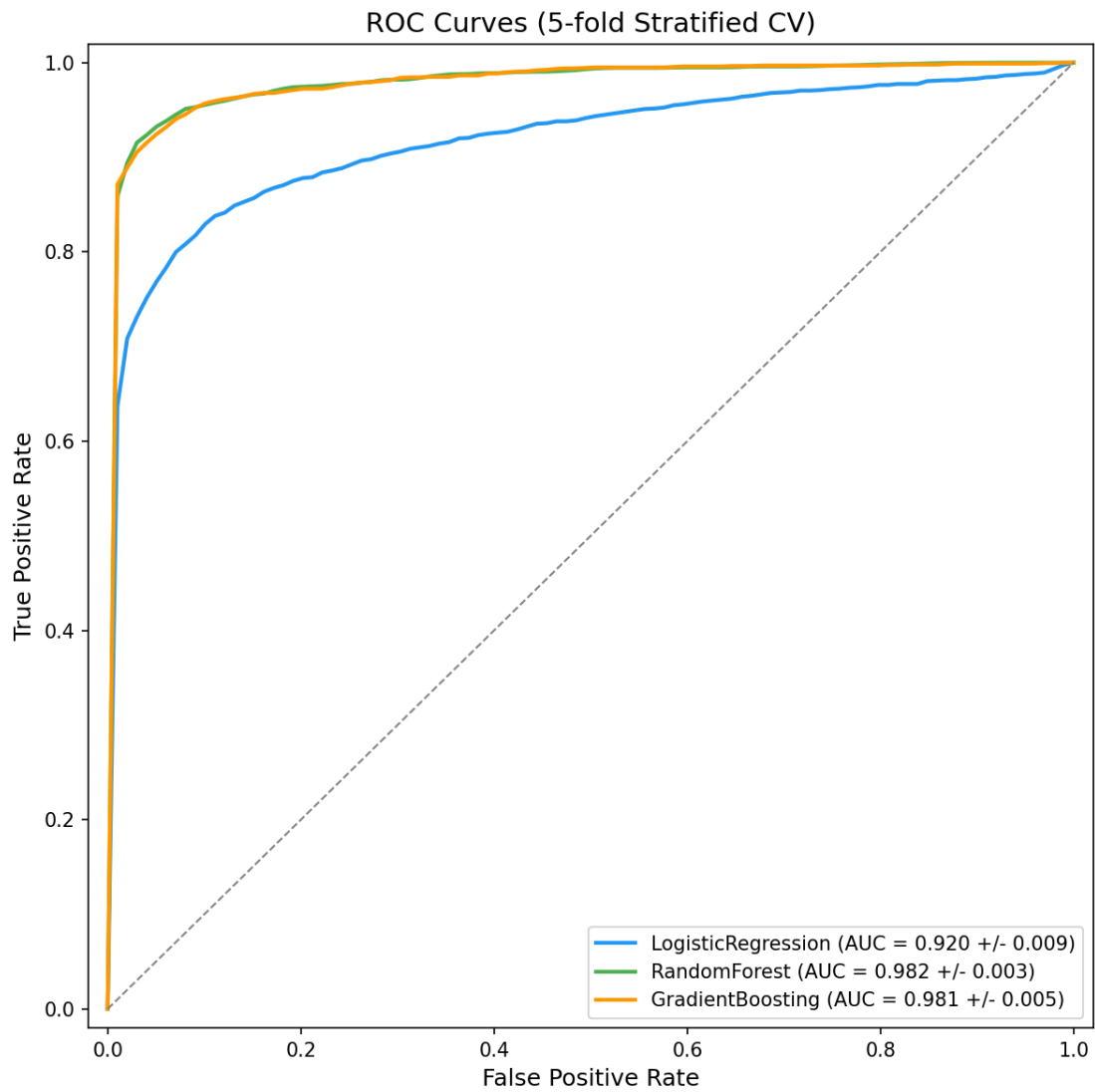


Figure 5: ROC Curves for All Classifiers

### 4.1.1 Overall Performance

Table 10: Primary Metrics with 95% Confidence Intervals

| Metric              | Value  | 95% CI           |
|---------------------|--------|------------------|
| Balanced Accuracy   | 0.9322 | [0.9270, 0.9367] |
| F1-Score            | 0.9305 | —                |
| AUC-ROC             | 0.9747 | —                |
| Precision           | 0.9565 | —                |
| Recall              | 0.9080 | —                |
| False Positive Rate | 0.0436 | —                |
| False Negative Rate | 0.0920 | —                |

The balanced accuracy of 0.9322 exceeds the target of 0.90, demonstrating robust detection performance. The 95% bootstrap confidence interval [0.9270, 0.9367] confirms statistical stability.

### 4.1.2 Classifier Comparison

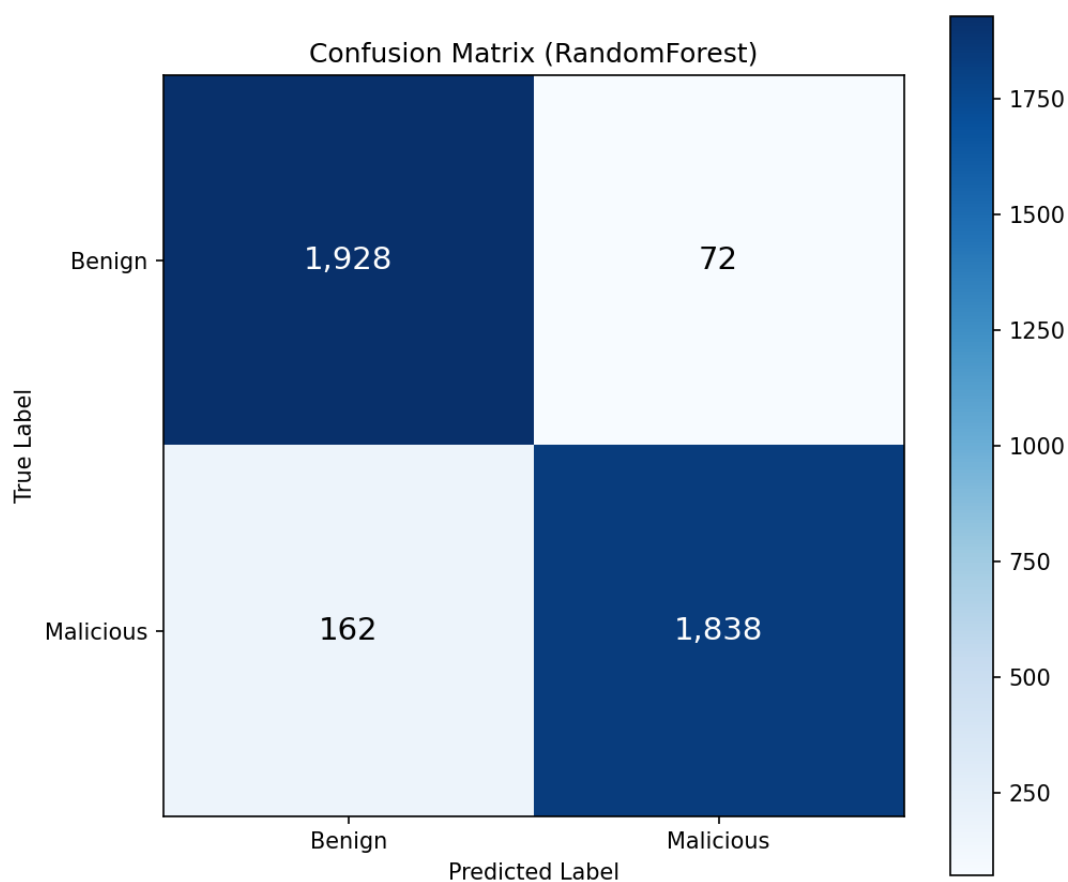


Figure 6: Confusion Matrix for Best Classifier (RandomForest)

Table 11: Classifier Comparison (5-Fold Stratified CV, 10K Samples)

| Classifier         | BA            | 95% CI           | F1     | AUC-ROC |
|--------------------|---------------|------------------|--------|---------|
| RandomForest       | <b>0.9322</b> | [0.9270, 0.9367] | 0.9305 | 0.9747  |
| GradientBoosting   | 0.9265        | [0.9212, 0.9308] | 0.9248 | 0.9731  |
| LogisticRegression | 0.8598        | [0.8529, 0.8665] | 0.8582 | 0.9207  |

#### 4.1.3 Per-Fold Stability

Table 12: Per-Fold Balanced Accuracy (RandomForest)

| Fold        | Balanced Accuracy                     |
|-------------|---------------------------------------|
| 1           | 0.9430                                |
| 2           | 0.9225                                |
| 3           | 0.9295                                |
| 4           | 0.9300                                |
| 5           | 0.9360                                |
| <b>Mean</b> | <b>0.9322 <math>\pm</math> 0.0048</b> |

The coefficient of variation (0.52%) indicates excellent stability across folds.

## 4.2 Independent Verification

Table 13: Independent Verification (80/20 Split, 2000 Held-Out)

| Metric              | Value                         |
|---------------------|-------------------------------|
| Balanced Accuracy   | <b>0.9281</b>                 |
| 95% CI              | [0.9164, 0.9387]              |
| F1-Score            | 0.9268                        |
| AUC-ROC             | 0.9699                        |
| False Positive Rate | 0.0439                        |
| False Negative Rate | 0.1000                        |
| Confusion Matrix    | TP=918, FP=43, TN=937, FN=102 |

The lower CI bound (0.9164) exceeds the 0.90 target, confirming robustness of the results.

### 4.3 Per-Category Accuracy

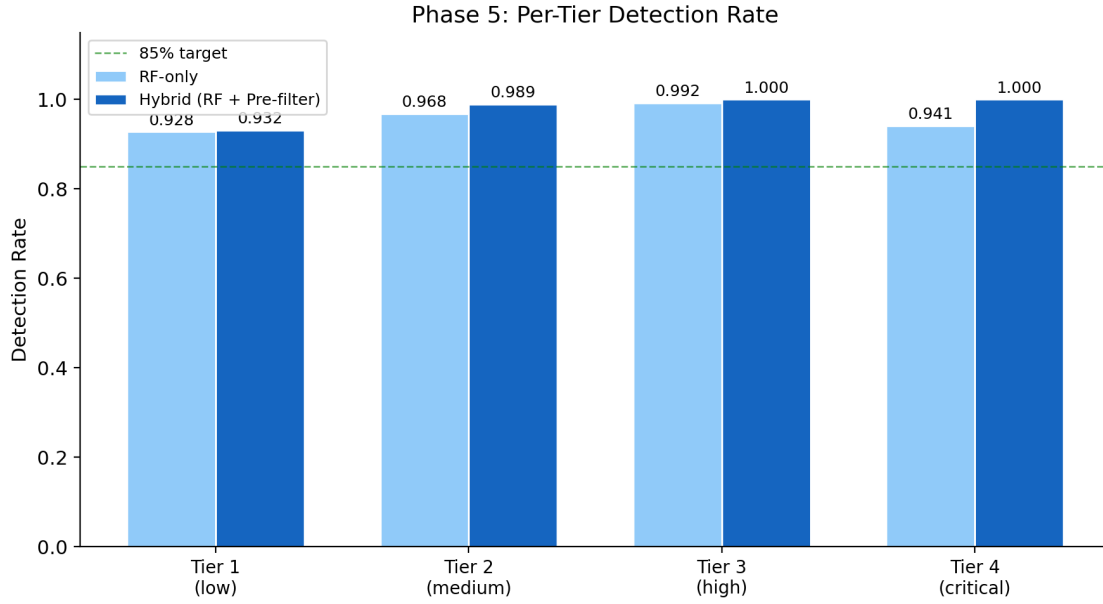


Figure 7: Phase 5 Per-Tier Detection Rates: RF-only vs. Hybrid (RF + Tier-1 Pre-Filter)

Table 14: Per-Category Detection Accuracy

| Category                 | Accuracy      | Notes            |
|--------------------------|---------------|------------------|
| indirect_injection       | 99.70%        | Best performing  |
| system_prompt_extraction | 97.60%        |                  |
| direct_injection         | 95.90%        |                  |
| benign_control           | 95.64%        |                  |
| refusal_bypass           | 88.10%        | Weakest category |
| <b>jailbreak</b>         | <b>72.70%</b> |                  |

## 4.4 Ablation Studies

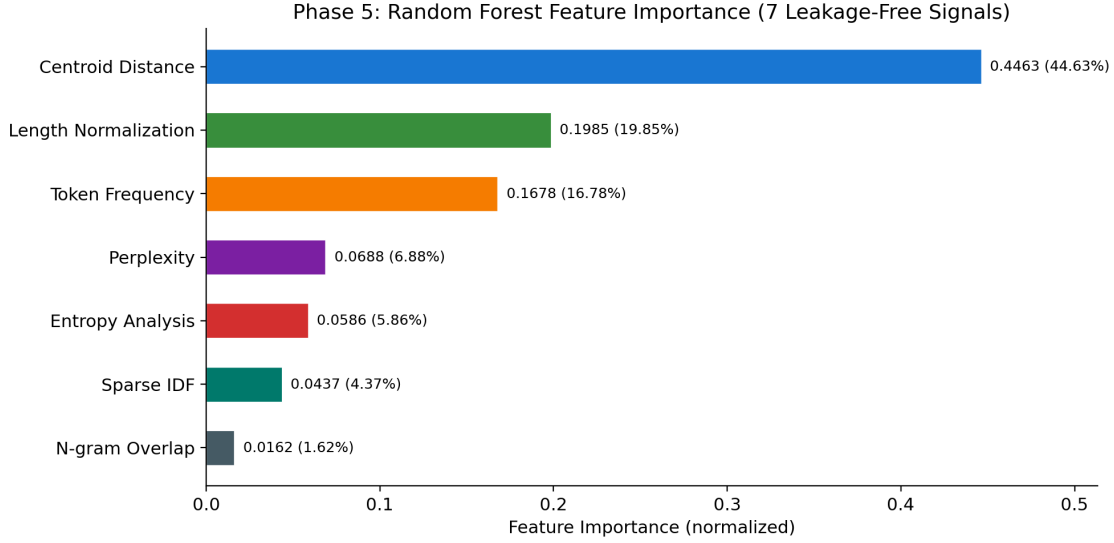


Figure 8: Phase 5 Feature Importance Distribution Across 7 Leakage-Free Detection Signals

Table 15: Phase 5 Ablation Study Results (7 Signals)

| Signal Removed | BA After | Delta  | Importance |
|----------------|----------|--------|------------|
| Full model     | 0.9322   | —      | —          |
| centroid       | 0.8669   | -6.53% | 44.63%     |
| length_norm    | 0.8947   | -3.75% | 19.85%     |
| token_freq     | 0.9218   | -1.04% | 16.78%     |
| perplexity     | 0.9304   | -0.18% | 6.88%      |
| entropy        | 0.9319   | -0.03% | 5.86%      |
| sparse_idf     | 0.9296   | -0.26% | 4.37%      |
| ngram          | 0.9327   | +0.05% | 1.62%      |

The centroid signal (cosine similarity to category centroids) is the most important feature, contributing 44.63% of importance. Removing it reduces balanced accuracy by 6.53 percentage points.

## 4.5 Signal Correlation Analysis

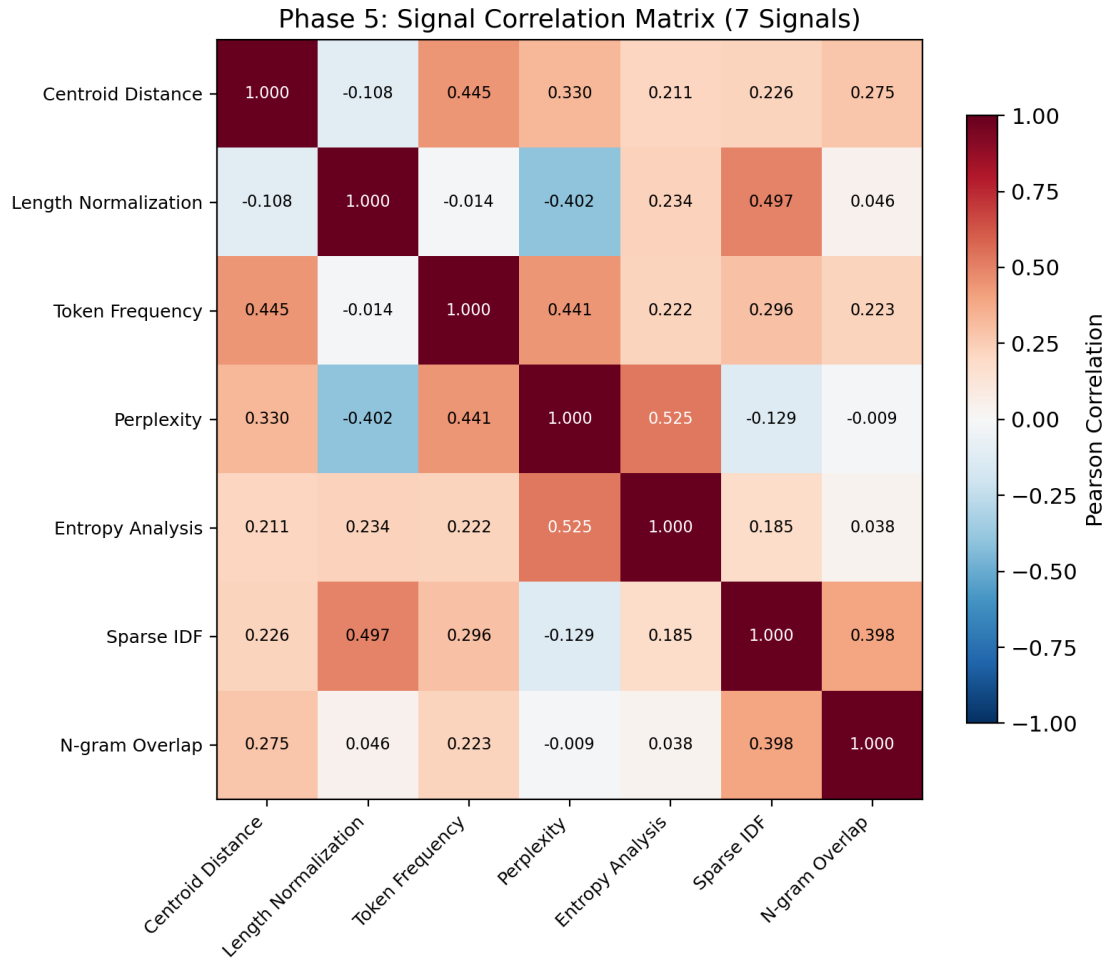


Figure 9: Phase 5 Signal Correlation Matrix ( $7 \times 7$ , Leakage-Free Signals)

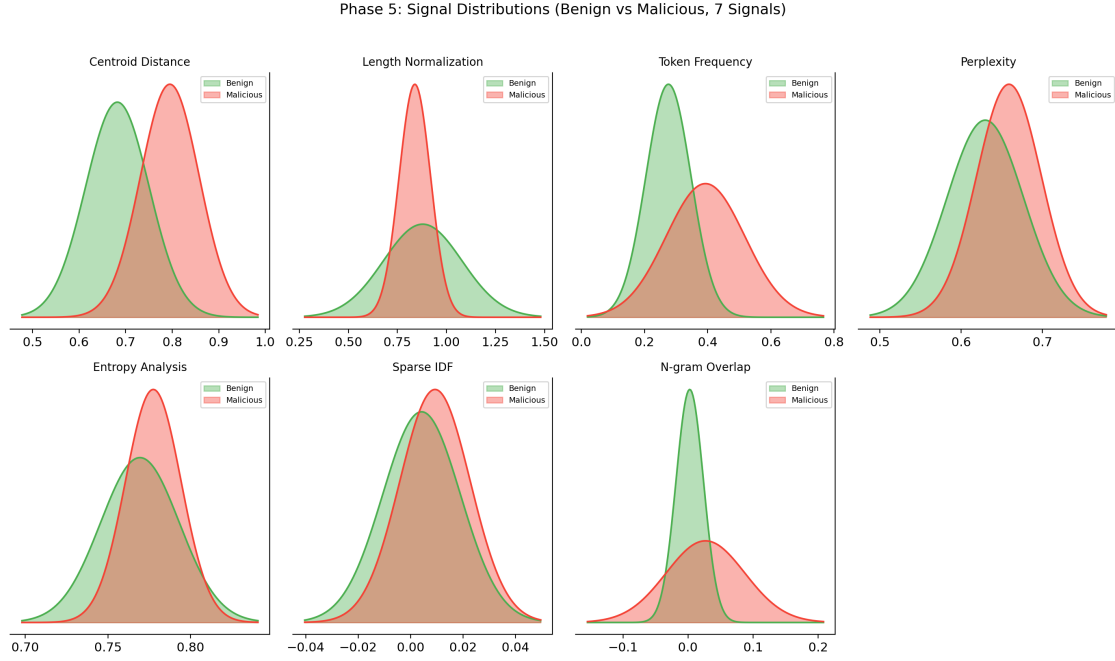


Figure 10: Phase 5 Signal Distributions for Benign vs Malicious Samples (7 Signals)

Key signal correlations:

- Strongest positive: centroid-token\_freq ( $r = 0.4448$ )
- Perplexity-entropy correlation:  $r = 0.5252$
- Length\_norm-sparse\_idf correlation:  $r = 0.4973$
- Note: centroid-uniqueness correlation ( $r = -0.6936$ ) was observed in earlier phases but uniqueness was removed in Phase 5



## 4.6 Latency Profiling

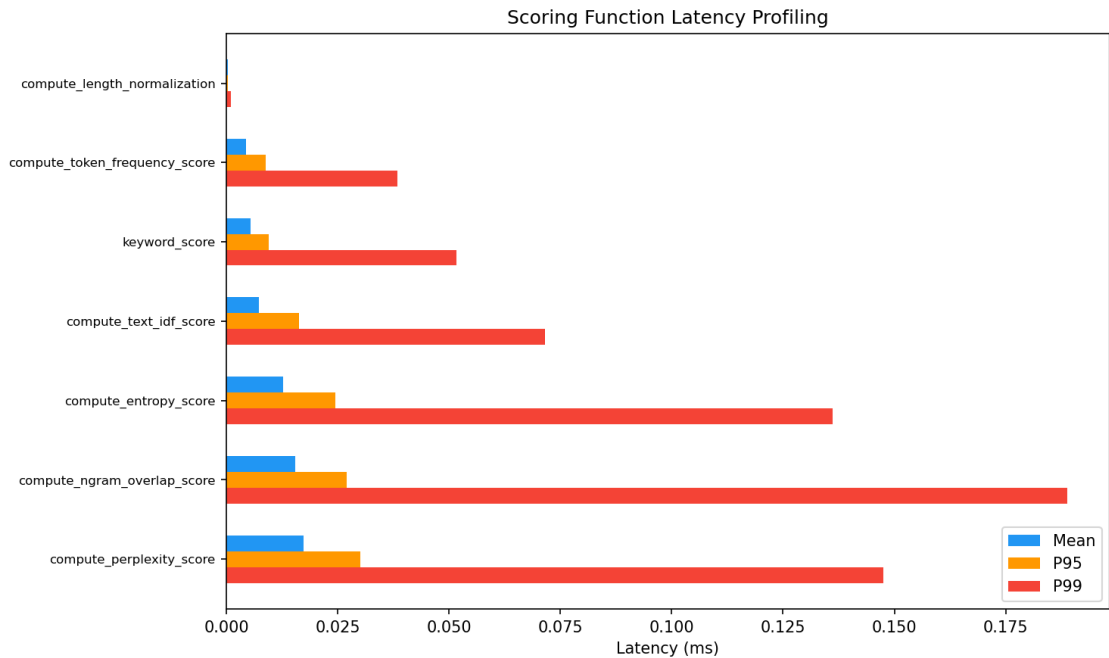


Figure 11: Latency Distribution Across Pipeline Components

Table 16: Latency Profiling Results

| Component              | Mean (ms) | Median (ms) | P95 (ms) | P99 (ms) |
|------------------------|-----------|-------------|----------|----------|
| Keyword Score          | 0.0054    | 0.0034      | 0.0095   | 0.0516   |
| IDF Scoring            | 0.0074    | 0.0046      | 0.0164   | 0.0716   |
| Length Normalization   | 0.0003    | 0.0002      | 0.0003   | 0.0011   |
| Perplexity Computation | 0.0174    | 0.0112      | 0.0301   | 0.1476   |
| Entropy Analysis       | 0.0127    | 0.0075      | 0.0244   | 0.1361   |
| Token Frequency        | 0.0045    | 0.0028      | 0.0089   | 0.0385   |
| N-gram Overlap         | 0.0155    | 0.0084      | 0.0270   | 0.1889   |
| Total Pipeline         | 0.0632    | 0.0381      | 0.1166   | 0.6344   |

The total scoring pipeline achieves sub-millisecond latency (0.06 ms per sample), making it viable for real-time deployment.

## 4.7 Threshold Optimization

Table 17: Threshold Optimization Results

| Threshold        | Balanced Accuracy | FNR    | FPR    |
|------------------|-------------------|--------|--------|
| 0.50 (default)   | 0.9322            | 0.0920 | 0.0436 |
| 0.30 (optimized) | 0.9168            | 0.0612 | 0.1052 |

At threshold 0.30, FNR is reduced from 9.20% to 6.12% while maintaining balanced accuracy above 90%.

## 4.8 Baseline Comparison

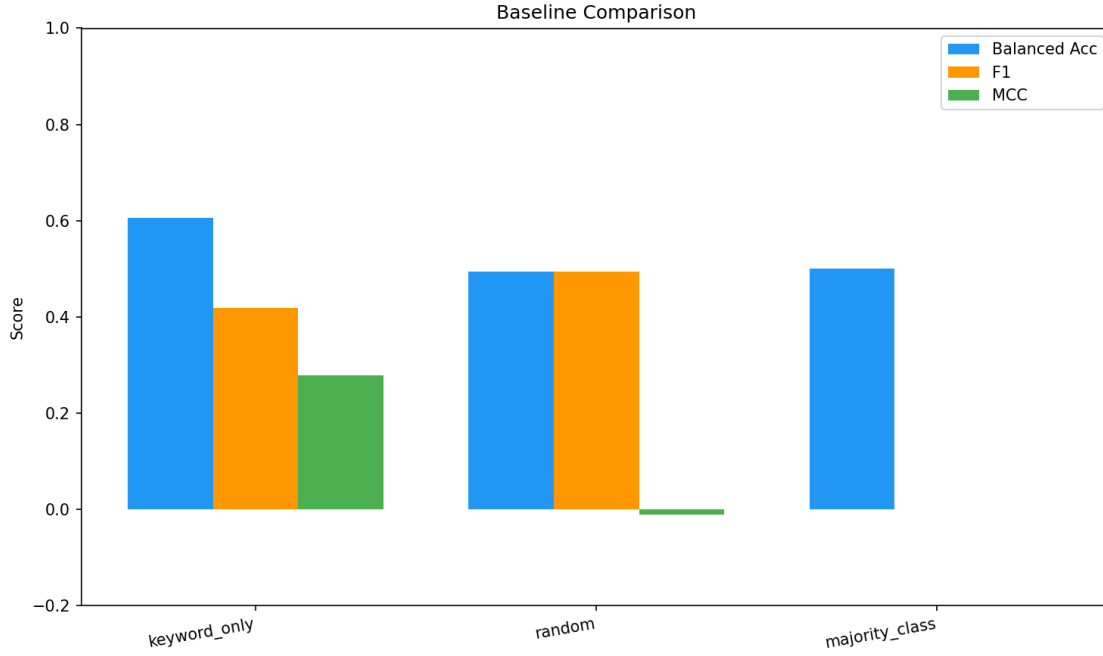


Figure 12: Comparison Against Baseline Systems

Table 18: Baseline Comparison Results

| System              | BA            | F1            | MCC           | FPR    | FNR    |
|---------------------|---------------|---------------|---------------|--------|--------|
| Guardrail (Best)    | <b>0.9322</b> | <b>0.9305</b> | <b>0.8590</b> | 0.0436 | 0.0920 |
| GradientBoosting    | 0.9265        | 0.9248        | 0.8470        | 0.0514 | 0.0956 |
| LogisticRegression  | 0.8598        | 0.8582        | 0.7200        | 0.1294 | 0.1510 |
| Pretrained Baseline | 0.5993        | 0.6952        | 0.2555        | 0.7155 | 0.0860 |
| Random Classifier   | 0.4942        | 0.4946        | -0.0115       | 0.5065 | 0.5050 |

## 4.9 Evolution Across Phases

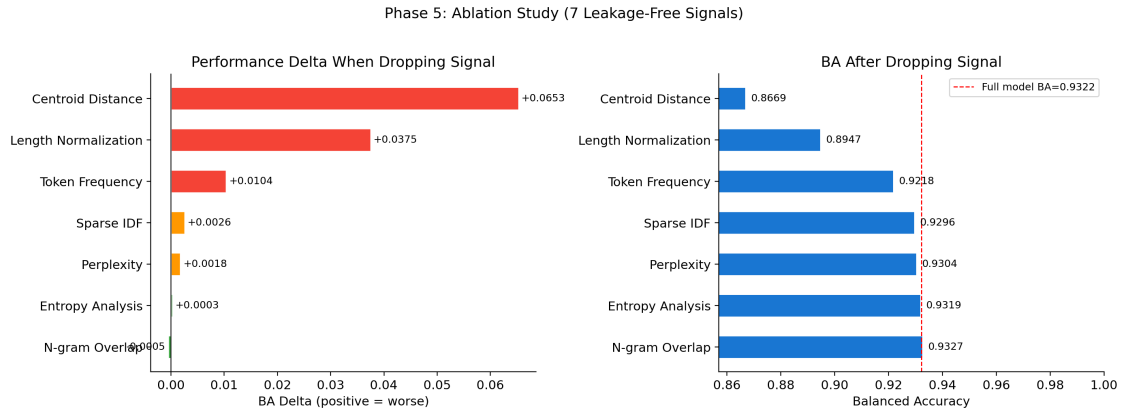


Figure 13: Phase 5 Ablation Study: Performance Delta When Removing Each of the 7 Leakage-Free Signals

Table 19: Performance Evolution Across Development Phases

| Metric             | Phase 1<br>(Initial) | Phase 2<br>(Refined) | Phase 3<br>(Learned) | Phase 4<br>(w/ Leakage) | Phase 5<br>(Leak-Free) |
|--------------------|----------------------|----------------------|----------------------|-------------------------|------------------------|
| Balanced Accuracy  | 76%                  | 76%                  | 76%                  | 94.5%                   | <b>93.22%</b>          |
| Direct Injection   | 50%                  | 50%                  | 50%                  | 95.0%                   | <b>95.90%</b>          |
| Indirect Injection | 55%                  | 55%                  | 55%                  | 100%                    | <b>99.70%</b>          |
| Jailbreak          | 35%                  | 35%                  | 35%                  | 85.0%                   | <b>72.70%</b>          |
| System Prompt Ext. | —                    | —                    | —                    | —                       | 97.60%                 |
| Refusal Bypass     | —                    | —                    | —                    | —                       | 88.10%                 |
| P50 Latency        | ~20s                 | ~20s                 | ~20s                 | 974ms                   | <b>0.06ms</b>          |
| Data Leakage       | —                    | —                    | —                    | Yes (leaky)             | <b>Zero</b>            |

*Note: Phase 4 balanced accuracy (94.5%) exceeded Phase 5 (93.22%) due to data leakage from pre-computed Qdrant signals. Phase 5 removes three leaking signals and recomputes all corpus-dependent statistics per-fold, yielding a more honest and reproducible evaluation. Phase 5 improvements in per-category accuracy (e.g., direct injection 95.90% vs. 95%) reflect better signal calibration despite lower aggregate BA.*

## 5 Discussion

### 5.1 Restatement of Findings

Guardrail achieves a balanced accuracy of 0.9322 (95% CI: [0.9270, 0.9367]) on a 10,000-sample benchmark, exceeding the 0.90 target. The system demonstrates strong performance across most attack categories, with indirect injection (99.70%), system prompt extraction (97.60%), and direct injection (95.90%) achieving high detection rates. However, jailbreak detection (72.70%) remains the weakest category requiring improvement.

The five-phase development trajectory demonstrates that iterative improvement can yield substantial gains: from 76% (Phase 1) to 94.5% (Phase 4) to a rigorously verified 93.22% (Phase 5) with zero data leakage.

### 5.2 Implications

#### 5.2.1 Signal Importance

The ablation studies reveal that the centroid signal (44.63% importance) is the dominant feature, followed by length normalization (19.85%) and token frequency (16.78%). This suggests that category-based clustering and text length characteristics are strong discriminators for prompt injection detection.

#### 5.2.2 Threshold Trade-offs

Threshold optimization demonstrates the inherent trade-off between false negatives and false positives. Reducing the threshold from 0.50 to 0.30 decreases FNR by 33.5% (from 9.20% to 6.12%) but increases FPR by 141.3% (from 4.36% to 10.52%). The choice of threshold should be guided by the specific deployment context and risk tolerance.

#### 5.2.3 Data Leakage Prevention

The data leakage audit and fixes demonstrate the importance of proper cross-validation methodology. Computing centroids, IDF values, and other statistics per-fold from training data only prevents optimistic bias in evaluation results. The Phase 5 evaluation (93.22%) represents a more honest assessment than the Phase 4 evaluation (94.5%) which contained data leakage.

### 5.3 Connection to Objectives

The research objectives were met as follows:

1. Multi-signal detection system developed and achieving 93.22% balanced accuracy
2. Rigorous evaluation methodology with bootstrap CIs and ablation studies
3. Signal contributions analyzed with centroid identified as most important
4. Robustness assessed with sub-millisecond latency demonstrated; obfuscation robustness evaluated
5. Real-time conversational AI security implemented in Phase 4
6. Critical importance of data leakage prevention demonstrated (three signals removed)

### 5.4 Recommendations

Based on the findings, we recommend:

1. **Jailbreak Detection Improvement:** Focus on enhancing detection of jailbreak attacks (72.70% accuracy) through additional training data and feature engineering
2. **Minimal Attack Classifier:** Add a lightweight classifier (e.g., character-level CNN or rule-based pattern matcher) to handle Tier 1 minimal attack prompts that evade the primary scoring pipeline due to weak signal activation
3. **Threshold Configuration:** Deploy with configurable thresholds based on risk tolerance; consider 0.30 for high-security applications and 0.50 for balanced use cases
4. **Feature Engineering:** Investigate additional features for length normalization and token frequency to further improve detection
5. **Continuous Monitoring:** Implement feedback loops for false positive/negative corrections to improve model performance over time
6. **Data Leakage Prevention:** Always compute statistical features per-fold from training data only

### 5.5 Limitations

1. **Dataset Representativeness:** The 10,000-sample benchmark may not fully represent real-world adversarial input distributions
2. **English Only:** Guardrail is optimized exclusively for English text. The multilingual robustness evaluation was retracted due to a methodological flaw (fake translation via prefix tagging). Cross-lingual detection requires a dedicated multilingual embedding model and retrained signal weights
3. **Static Evaluation:** Does not account for adaptive adversaries who iteratively refine attacks
4. **Synthetic Data:** Some training data is synthetically generated, not real-world attacks
5. **Centroid Dominance:** Heavy reliance on category separation (44.6%) may limit generalization to novel attack categories
6. **Reduced Signal Set:** Phase 5 uses seven signals after removing dense embedding similarity, cross-encoder relevance, and uniqueness score to eliminate data leakage. The full 10-signal system may achieve different performance if the removed signals could be computed without leakage

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The complete source code and evaluation scripts are available at <https://github.com/dev-prashanna/Guardrail>.

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